

ELEC 221 Lecture 04

The convolution integral; the impulse response and system properties

Tuesday 22 September 2026

Announcements

- Tutorial assignment 1 due Friday 23:59
- Quiz 2 today

The convolution sum

We expressed signals as weighted sums of impulses

If we know what an LTI system does to a unit impulse (i.e., the impulse response $h[n]$), we know what it does to any other signal:

This is the **convolution sum**. Today we will see the **convolution integral** in continuous time.

Today

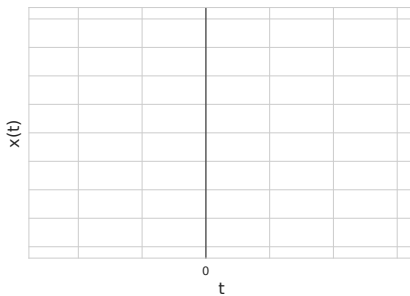
Learning outcomes:

- Define the CT unit impulse and step functions
- Define the convolution integral and use it to compute the output of a system
- Use the impulse response to determine whether a system is stable, causal, memoryless, or invertible

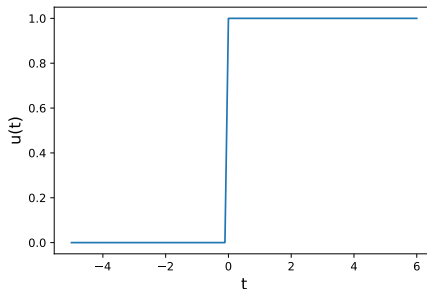
If we have time:

- Use the convolution sum/integral to show that complex exponentials are eigenfunctions of LTI systems

The CT unit impulse



The CT unit step



Just like in DT, the unit impulse and step are related:

The convolution integral

The CT analogue of convolution sum is the **convolution integral**.

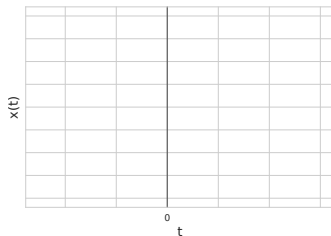
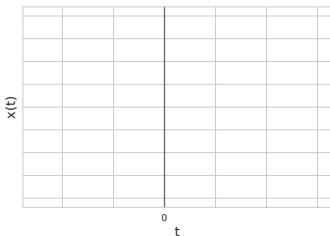
where $h(t)$ is the **CT impulse response**.

It has the same properties (commutative, associative, distributive).

Example: convolution

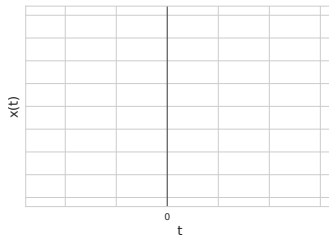
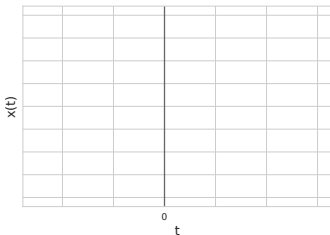
(Oppenheim Ex. 2.6 Var.) Consider system with impulse response

What is the output of the system for the input signal



Example: convolution

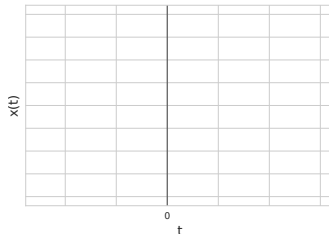
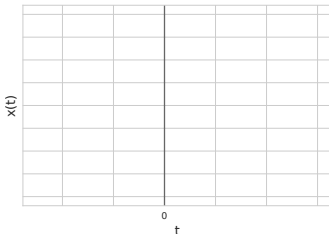
Example: convolution



Exercise: convolution

(Oppenheim 2.8) Consider system with impulse response

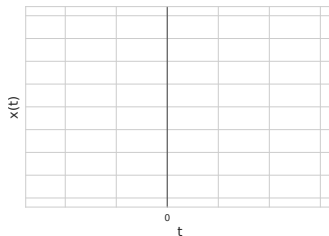
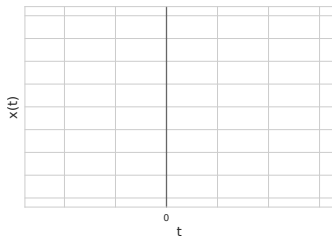
What is the output of the system for the input signal



Exercise: convolution

Direct integration:

Visual intuition:



Impulse response and analysis of LTI systems

To reiterate: the convolution sum

and convolution integral

show that as long as we know how a system responds to a unit impulse, we can determine its response to any other signal.

Knowledge of the impulse response also allows us to determine key properties of systems such as causality, invertibility, and stability.

Impulse response and memory

A system is memoryless if the output depends only on the input at the same time. This implies $h[n] = 0$ for $n \neq 0$, meaning

In CT, $h(t) = K\delta(t)$.

Impulse response and invertibility

If a system is invertible, it has an inverse system.
Suppose impulse response of a system is $h(t)$. Then

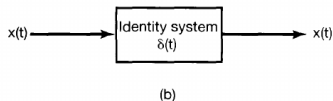
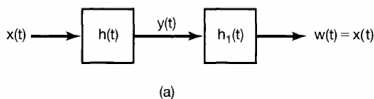


Figure 2.26 Concept of an inverse system for continuous-time LTI systems. The system with impulse response $h_1(t)$ is the inverse of the system with impulse response $h(t)$ if $h(t) * h_1(t) = \delta(t)$.

(Analogous for DT case. We will see this later in the course.)

Impulse response and stability

Suppose we have a signal $x(t)$ with a bounded input, $|x(t)| \leq B$.
If the system is stable, the output should be bounded.

Impulse response and stability

As long as

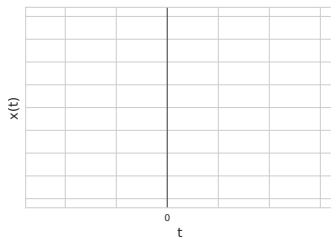
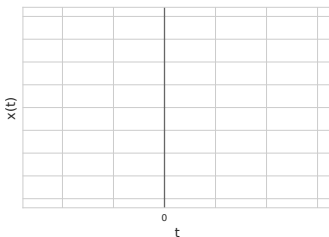
(i.e., $h(t)$ is *absolutely integrable*), the system is stable.

In DT, $h[n]$ must be *absolutely summable* to ensure stability:

Example/exercise: stability

Consider systems A and B with impulse responses

Are they stable?



Example/exercise: stability

Impulse response and causality

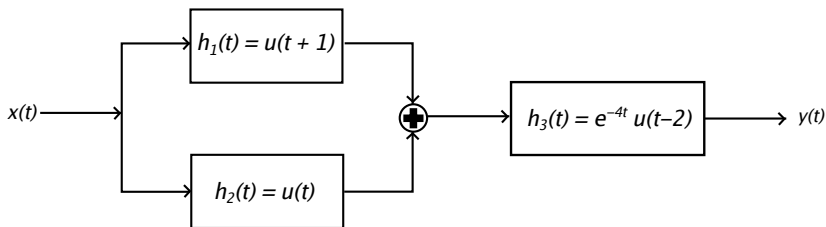
Recall definition of causal signal and consider the convolution sum:

What properties does $h[n]$ need to have for system to be causal?

(Analogous holds for CT systems)

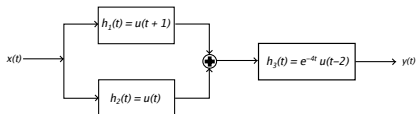
Final exercise

Consider the following combination of systems:

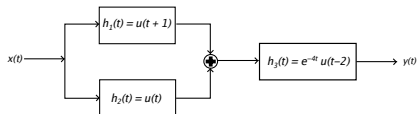


Is this system causal and/or stable?

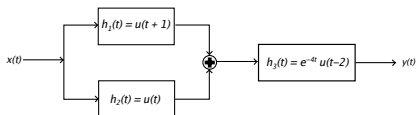
Final exercise



Final exercise



Final exercise

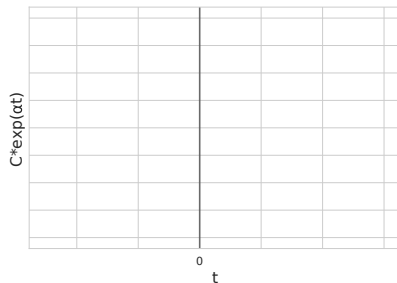
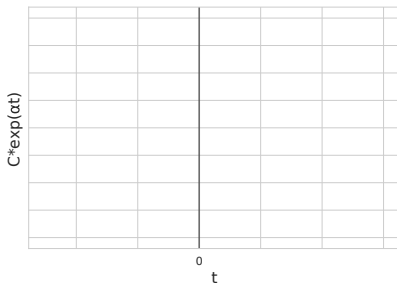


Review: complex exponential functions

Most general form:

where α can be real or complex.

Case: both C and α are real-valued.



Review: complex exponential functions

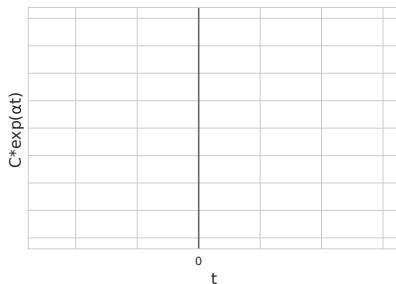
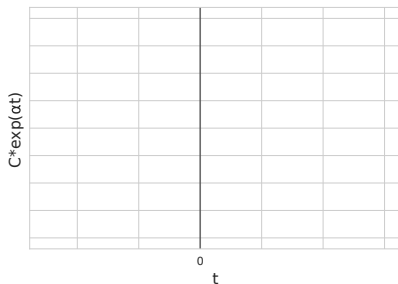
Case: α is complex. Recall can write any complex z in two ways:

Euler's relation allows us to rewrite the second form:

As a result,

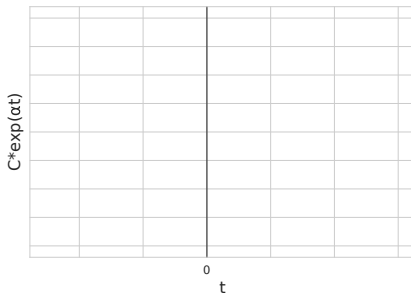
Review: complex exponential functions

Case: α is a complex number $\alpha = j\omega$. Then,



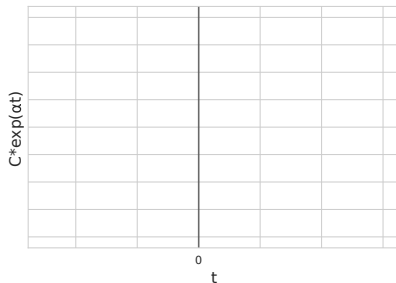
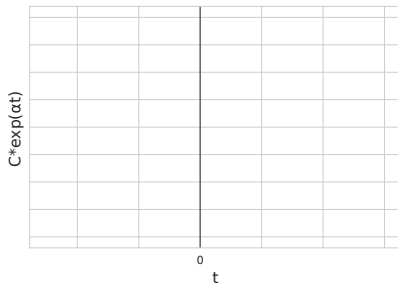
Review: complex exponential functions

$x(t) = Ce^{j\omega t}$ is periodic:



Review: complex exponential functions

Case: α is a complex number $\alpha = r + j\omega$. Then,



LTI systems and complex exponential functions

Recall the convolution integral:

What happens when our signal $x(t)$ is a complex exponential function?

LTI systems and complex exponential functions

Write $x(t) = e^{st}$. Then:

LTI systems and complex exponential functions

To summarize:

Complex exponentials are **eigenfunctions** of LTI systems.

$H(s)$ is called the **system function**, or *frequency response*, of an LTI system.

...so what?

Recall that LTI system sends superpositions of inputs to superpositions of outputs. If $x(t) \rightarrow y(t)$,

...so what?

If all the $x_i(t)$ are complex exponential functions and

then

The response of LTI systems to superpositions of complex exponential signals is a superposition **of those same signals**.

How can we express arbitrary signals using complex exponentials?

For next time

Content:

- (Complex) exponential signals
- The continuous-time Fourier series representation
- CT Fourier series representation and properties

Action items:

- ① Tutorial Assignment 1 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 1.4, 2.2-2.3
- Practice problems: 2.8-2.12, 2.14-16, 2.22, 2.28, 2.29
- For next class: Oppenheim 1.3, 3.0-3.5